Lab 7: Fuzzing for Input Validation Bugs

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Environment: Ubuntu 22.04.5 LTS (Linux 6.1.189 x86_64), Python 3.10.12,
hypothesis 6.168.5, pytest 9.2.0 (installed with pip in a virtual environment).
Repo files: photo_print_order.py and test_photo_print_order.py
(the handout calls them processor.py and test_processor.py).

Step 1: Install dependencies

pip install hypothesis and pip install pytest (in a fresh virtual environment).

```
$ pip install hypothesis
Collecting hypothesis
  Downloading hypothesis-6.168.5-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (1.2 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.2/1.2 MB 38.0 MB/s eta 0:00:00
Collecting sortedcontainers<3.0.0,>=2.1.0
  Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl (29 kB)
Collecting exceptiongroup>=1.0.0
  Downloading exceptiongroup-1.3.1-py3-none-any.whl (16 kB)
Collecting typing-extensions>=4.6.0
  Downloading typing_extensions-4.16.0-py3-none-any.whl (45 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 45.6/45.6 KB 317.5 MB/s eta 0:00:00
Installing collected packages: sortedcontainers, typing-extensions, exceptiongroup, hypothesis
Successfully installed exceptiongroup-1.3.1 hypothesis-6.168.5 sortedcontainers-2.4.0 typing-extensions-4.16.0
$ pip install pytest
Collecting pytest
  Downloading pytest-9.2.0-py3-none-any.whl (411 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 411.8/411.8 KB 20.7 MB/s eta 0:00:00
Collecting pygments>=2.15
  Downloading pygments-2.21.0-py3-none-any.whl (1.3 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.3/1.3 MB 112.4 MB/s eta 0:00:00
Collecting iniconfig>=2
  Downloading iniconfig-2.3.1-py3-none-any.whl (7.6 kB)
Collecting tomli>=2
  Downloading tomli-2.5.0-py3-none-any.whl (14 kB)
Collecting pluggy<2,>=1.6
  Downloading pluggy-1.7.0-py3-none-any.whl (27 kB)
Requirement already satisfied: exceptiongroup>=1 in /tmp/v2/lib/python3.10/site-packages (from pytest) (1.3.1)
Collecting packaging>=24
  Downloading packaging-26.3-py3-none-any.whl (129 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 130.0/130.0 KB 601.0 MB/s eta 0:00:00
Requirement already satisfied: typing-extensions>=4.6.0 in /tmp/v2/lib/python3.10/site-packages (from exceptiongroup>=1->
Installing collected packages: tomli, pygments, pluggy, packaging, iniconfig, pytest
Successfully installed iniconfig-2.3.1 packaging-26.3 pluggy-1.7.0 pygments-2.21.0 pytest-9.2.0 tomli-2.5.0
$ pip list | grep -i -E "hypothesis|pytest"
hypothesis        6.168.5
pytest            9.2.0
```

Step 2: Run the buggy script

python photo_print_order.py. Valid input works; malformed input crashes with ValueError.

```
$ python photo_print_order.py   # valid input
==== Running parse_order ====
Enter print size:copies (e.g. 4x6:25): Size and Copies: ('4X6', 25)

==== Running file_size_average ====
Enter file sizes in MB as CSV (e.g. 4,8,12): Average File Size: 8

==== Running payable_amount ====
Enter amount|discount (e.g. 450|50): Payable Amount: 481.0

$ python photo_print_order.py   # invalid input
==== Running parse_order ====
Enter print size:copies (e.g. 4x6:25): Traceback (most recent call last):
  File "/tmp/r/Lab 7/photo_print_order.py", line 31, in <module>
    main()
  File "/tmp/r/Lab 7/photo_print_order.py", line 17, in main
    result = parse_order(user_input)
  File "/tmp/r/Lab 7/photo_print_order.py", line 2, in parse_order
    size, copies = text.split(':')
ValueError: not enough values to unpack (expected 2, got 1)
```

Step 4: pytest on the buggy code (tests fail)

Hypothesis finds crashing inputs for all three functions (empty string, ":", None, ...).

```
$ pytest test_photo_print_order.py
============================= test session starts ==============================
platform linux -- Python 3.10.12, pytest-9.2.0, pluggy-1.7.0
rootdir: /tmp/r/Lab 7
plugins: hypothesis-6.168.5
collected 3 items

test_photo_print_order.py FFF                                            [100%]

=================================== FAILURES ===================================
_____________________________ test_parse_order_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 5, in test_parse_order_no_crash
  |     def test_parse_order_no_crash(s):
  | exceptiongroup.ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 7, in test_parse_order_no_crash
    |     parse_order(s)
    |   File "/tmp/r/Lab 7/photo_print_order.py", line 3, in parse_order
    |     return size.strip().upper(), int(copies)
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 9, in test_parse_order_no_crash
    |     assert False, f"parse_order crashed with: {e!r}"
    | AssertionError: parse_order crashed with: ValueError("invalid literal for int() with base 10: ''")
    | assert False
    | Failing test case: test_parse_order_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 7, in test_parse_order_no_crash
    |     parse_order(s)
    |   File "/tmp/r/Lab 7/photo_print_order.py", line 2, in parse_order
    |     size, copies = text.split(':')
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 9, in test_parse_order_no_crash
    |     assert False, f"parse_order crashed with: {e!r}"
    | AssertionError: parse_order crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
    | assert False
    | Failing test case: test_parse_order_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 7, in test_parse_order_no_crash
    |     parse_order(s)
    |   File "/tmp/r/Lab 7/photo_print_order.py", line 2, in parse_order
    |     size, copies = text.split(':')
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 9, in test_parse_order_no_crash
    |     assert False, f"parse_order crashed with: {e!r}"
    | AssertionError: parse_order crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_parse_order_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________ test_file_size_average_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 12, in test_file_size_average_no_crash
  |     def test_file_size_average_no_crash(s):
  | exceptiongroup.ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 14, in test_file_size_average_no_crash
    |     file_size_average(s)
```

(output continued)

```
    |   File "/tmp/r/Lab 7/photo_print_order.py", line 6, in file_size_average
    |     values = [int(r) for r in sizes_csv.split(',')]
    |   File "/tmp/r/Lab 7/photo_print_order.py", line 6, in <listcomp>
    |     values = [int(r) for r in sizes_csv.split(',')]
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 16, in test_file_size_average_no_crash
    |     assert False, f"file_size_average crashed with: {e!r}"
    | AssertionError: file_size_average crashed with: ValueError("invalid literal for int() with base 10: ''")
    | assert False
    | Failing test case: test_file_size_average_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 14, in test_file_size_average_no_crash
    |     file_size_average(s)
    |   File "/tmp/r/Lab 7/photo_print_order.py", line 6, in file_size_average
    |     values = [int(r) for r in sizes_csv.split(',')]
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 16, in test_file_size_average_no_crash
    |     assert False, f"file_size_average crashed with: {e!r}"
    | AssertionError: file_size_average crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_file_size_average_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________ test_payable_amount_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 19, in test_payable_amount_no_crash
  |     def test_payable_amount_no_crash(s):
  | exceptiongroup.ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 21, in test_payable_amount_no_crash
    |     payable_amount(s)
    |   File "/tmp/r/Lab 7/photo_print_order.py", line 11, in payable_amount
    |     total = float(amount) * 1.18 - float(discount)
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 23, in test_payable_amount_no_crash
    |     assert False, f"payable_amount crashed with: {e!r}"
    | AssertionError: payable_amount crashed with: ValueError("could not convert string to float: ''")
    | assert False
    | Failing test case: test_payable_amount_no_crash(
    |     s='|',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 21, in test_payable_amount_no_crash
    |     payable_amount(s)
    |   File "/tmp/r/Lab 7/photo_print_order.py", line 10, in payable_amount
    |     amount, discount = text.split('|')
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 23, in test_payable_amount_no_crash
    |     assert False, f"payable_amount crashed with: {e!r}"
    | AssertionError: payable_amount crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
    | assert False
    | Failing test case: test_payable_amount_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 21, in test_payable_amount_no_crash
    |     payable_amount(s)
    |   File "/tmp/r/Lab 7/photo_print_order.py", line 10, in payable_amount
```

(output continued)

```
    |     amount, discount = text.split('|')
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/tmp/r/Lab 7/test_photo_print_order.py", line 23, in test_payable_amount_no_crash
    |     assert False, f"payable_amount crashed with: {e!r}"
    | AssertionError: payable_amount crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_payable_amount_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
========================== short test summary info ===========================
FAILED test_photo_print_order.py::test_parse_order_no_crash - exceptiongroup....
FAILED test_photo_print_order.py::test_file_size_average_no_crash - exception...
FAILED test_photo_print_order.py::test_payable_amount_no_crash - exceptiongro...
============================= 3 failed in 4.05s ==============================
```

Step 5: Run the fixed script

Valid input gives the same results; invalid input prints "Invalid input" instead of crashing.

```
$ python photo_print_order.py   # valid input
==== Running parse_order ====
Enter print size:copies (e.g. 4x6:25): Size and Copies: ('4X6', 25)

==== Running file_size_average ====
Enter file sizes in MB as CSV (e.g. 4,8,12): Average File Size: 8

==== Running payable_amount ====
Enter amount|discount (e.g. 450|50): Payable Amount: 481.0

$ python photo_print_order.py   # invalid input
==== Running parse_order ====
Enter print size:copies (e.g. 4x6:25): Size and Copies: Invalid input

==== Running file_size_average ====
Enter file sizes in MB as CSV (e.g. 4,8,12): Average File Size: Invalid input

==== Running payable_amount ====
Enter amount|discount (e.g. 450|50): Payable Amount: Invalid input
```

Step 6: pytest on the fixed code

All 7 tests pass (3 no-crash fuzz tests, 3 result-property fuzz tests, 1 example test).

```
$ pytest test_photo_print_order.py
============================= test session starts ==============================
platform linux -- Python 3.10.12, pytest-9.2.0, pluggy-1.7.0
rootdir: /tmp/r/Lab 7
plugins: hypothesis-6.168.5
collected 7 items

test_photo_print_order.py .......                                        [100%]

============================== 7 passed in 0.45s ===============================
```

Bugs found and fixed

- None input -> AttributeError (all 3 functions): type-checked, returns None.
- Missing/extra ":" or "|" -> ValueError on unpacking: parts are counted first.
- Empty or non-numeric fields (":", "", "a,b") -> ValueError: caught, returns None.
- Empty CSV and negative sizes rejected.
- float "nan", "inf", "1e999" slipped through: non-finite values rejected.
- Negative copies, amount or discount rejected.

Invalid input returns None; main() prints "Invalid input".